

Cem Yildiz

Mathematics and Computer Science Student | Applied ML and Data Analysis

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PROFILE

Mathematics and Computer Science student applying probability and statistics in Python data analysis and ML projects. Interested in careful model evaluation and building stronger ML/DL foundations.

EDUCATION

Eskisehir Osmangazi University - Mathematics and Computer Science studies | 2025-Present

Transferred after first-year Mathematics studies at Nevsehir Haci Bektas Veli University (2024-2025).

Relevant coursework: Probability and Statistics, Linear Algebra, Discrete Mathematics, Artificial Intelligence.

SELECTED PROJECTS

AI-assisted Fashion-MNIST Study | NumPy, MLP, computer vision

Evaluated a 128–64 two-hidden-layer ReLU MLP against a linear softmax baseline on 54,000 training and the fixed 10,000-image test split. The MLP reached 87.03% accuracy (macro-F1 0.8692) vs. 83.56% (0.8359) for the baseline.

Descriptive result from one seed and split; AI-assisted work. Repository:

github.com/cemyildizcy/fashion-mnist-numpy-capacity

AI-assisted Bike Demand Evaluation | Python, NumPy, pandas | UCI Bike Sharing (CC BY 4.0)

Compared a log-count ridge model with a weekday/hour seasonal baseline on 17,379 hourly records, using chronological 70/15/15 splits and train-only preprocessing. The calendar plus observed-weather model reached final-test MAE 77.79 (RMSE 112.63; $n=2,607$), versus baseline MAE 103.54. Weather is retrospective; this covers one city and two years.

Repository: github.com/cemyildizcy/bike-demand-temporal-ml

World Cup Match Simulator | Python, Streamlit

Built a Poisson score simulation and Monte Carlo tournament workflow, with a dashboard for exploring match-score and tournament outcome distributions. This is an independent educational simulation, not an official forecast.

Near-Earth Object Risk Analysis | Python, NASA NeoWs API

Built a workflow to fetch and clean close-approach records, combine diameter, velocity, and miss distance into a heuristic score, and produce six exploratory plots. The score is not a validated impact-probability model.

TECHNICAL SKILLS

Python, SQL, pandas, NumPy, scikit-learn, XGBoost, Streamlit, Matplotlib, Seaborn, Git

Exploratory data analysis, feature engineering, probability and statistics, Monte Carlo simulation, temporal model evaluation